

Hawkeye EQMS · Audit Request / RFQ (marketplace audit flow) · Feature Guide

Pharma-EQMS comparison · click-by-click guide · personas + lifecycle walkthrough · AI map · test results.

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1. Module overview

Module	Audit Request / RFQ (marketplace audit flow)
Pharma purpose	Buyer posts an audit RFQ. Multiple auditor orgs submit quotes. Buyer shortlists + awards. Awarded quote converts into a full audit request that enters the 8-phase audit lifecycle.
Compliance basis	ICH Q10 §2.7 · ISO 19011 §5.3 · contract audit governance
Hawkeye routes	/rfqs, /rfqs/[id], /request-audit (frontend) · /api/rfqs, /api/audit-requests (backend)
Hawkeye model file	backend/src/models/auditRfqModel.js · auditRfqQuoteModel.js · auditRfqThreadModel.js
Tenant module flag	modules.RFQ

RFQs follow DRAFT → PUBLISHED → IN_QA → QUOTES_RECEIVED → SHORTLISTED → AWARDED → CONVERTED. Quote rows follow DRAFT → SUBMITTED → REVISED / WITHDRAWN / ACCEPTED / REJECTED. An awarded quote auto-creates an audit-requests-master row that enters the 8-phase audit lifecycle.

2. Standard pharma EQMS vs Hawkeye — gap analysis

Each row is a capability a pharma QA team would expect from an EQMS deviation module. Hawkeye column shows what's actually shipped. **MET** = parity, **PARTIAL** = ships but with caveats, **GAP** = not implemented yet.

Pharma expectation	Hawkeye implementation	Status	Note
RFQ lifecycle (DRAFT → AWARDED → CONVERTED) —	9-state enum.	MET	
Per-auditor quote lifecycle (DRAFT → SUBMITTED → ACCEPTED/REJECTED) —	6-state enum on auditRfqQuoteModel.	MET	
Auto-numbered RFQ (RFQ-XXXXXX) GMP	Pre-save generator.	MET	
Invitations to specific auditor orgs —	invitedAuditors[] subdocument with status (INVITED / DECLINED / ACCEPTED_VIEW).	MET	
Q&A threads between buyer + auditor —	auditRfqThreadModel.	MET	
Quote line items + totals + proposed schedule —	lineItems[] + totals { subtotal, tax, total } + proposedSchedule.	MET	
Award + auto-convert to audit request —	POST /:id/award sets awardedQuoteld; downstream auto-creates audit-requests-master.	MET	
Closing date + auto-EXPIRE —	Vercel cron (03:30 UTC daily) calls /api/quality/scan-expirations → flips PUBLISHED/QUOTES_RECEIVED RFQs past closingAt to EXPIRED.	MET	

3. Personas + role mapping

Login password for every persona: `EqmsDemo@2026` . Roles drive which API endpoints respond — see Hawkeye `roleMiddleware.permit()` .

Persona	Email	Responsibilities in this module	Lifecycle states they touch
Audit Program Mgr Priya buyer	<code>audit.program@novex-pharma.demo</code>	Drafts RFQ, publishes, invites auditor orgs, shortlists, awards.	DRAFT PUBLISHED SHORTLISTED AWARDED

Persona	Email	Responsibilities in this module	Lifecycle states they touch
Auditor (external auditor org) auditor	(auditor-side login)	Receives invite, submits quote, revises after Q&A.	(quote) DRAFT SUBMITTED REVISED
Audit Lead Maria (auditor) auditor	audit.lead@novex-pharma.demo	Internal-tenant equivalent — submits quotes from Maria's auditor org.	(quote) DRAFT SUBMITTED

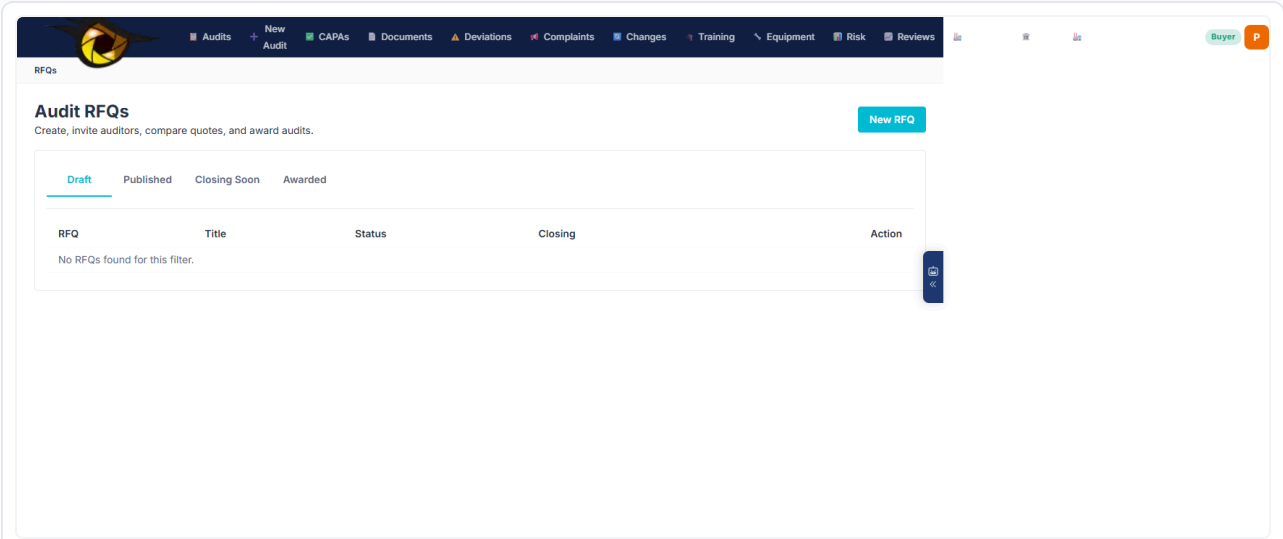
4. Feature catalogue (click-by-click)

4.1 RFQ register

What it does	Lists all RFQs with status + closingAt + invited-auditors count.
Menu / route	/rfqs
Available to roles	buyer · admin
Backend API	GET /api/rfqs

CLICK-BY-CLICK

- NAVIGATE** Click 'RFQs' in the top nav
→ Page renders



state-screens/rfq-list.png

4.2 + New RFQ

What it does	Buyer drafts an RFQ.
Menu / route	/rfqs · top-right '+ New RFQ'
Available to roles	buyer

Backend API

POST /api/rfqs

CLICK-BY-CLICK

1. **CLICK** Click '+ New RFQ'
→ Form opens
2. **TYPE** Fill title
→ required
3. **CLICK** Pick supplierOrgId + siteld
→ required
4. **CLICK** Pick productIds[]
5. **CLICK** Pick closingAt (deadline for quotes)
→ required
6. **CLICK** Click 'Save draft'
→ Status=DRAFT · rfqNumber=RFQ-XXXXXX

4.3 Publish

What it does	Move DRAFT → PUBLISHED so invited auditors can see it.
Menu / route	RFQ row · 'Publish' button
Available to roles	buyer
Backend API	POST /api/rfqs/:id/publish

CLICK-BY-CLICK

1. **CLICK** Click 'Publish'
→ Status flips

4.4 Invite auditors

What it does	Add specific auditor orgs to the invited list.
Menu / route	RFQ detail · 'Invite Auditors' button
Available to roles	buyer
Backend API	POST /api/rfqs/:id/invite

CLICK-BY-CLICK

1. **CLICK** Click 'Invite Auditors'
→ Drawer with auditor org picker
2. **CLICK** Pick auditorOrgIds[]
3. **CLICK** Click 'Send invitations'
→ Each gets status=INVITED in invitedAuditors[]; notification sent

4.5 Submit quote (auditor)

What it does	Auditor submits a quote with line items + totals + proposed schedule.
Menu / route	RFQ detail (auditor view) · '+ Submit Quote' button
Available to roles	auditor
Backend API	POST /api/rfqs/:id/quotes

CLICK-BY-CLICK

- CLICK** Click '+ Submit Quote'
→ Form opens
- TYPE** Fill lineItems[] (each: description + qty + unitPrice)
- TYPE** Fill totals.subtotal + tax + total
- TYPE** Fill proposedSchedule + assumptionsText
- CLICK** Click 'Submit'
→ Quote row · status=SUBMITTED · RFQ status auto → QUOTES_RECEIVED

4.6 Shortlist + Award

What it does	Buyer narrows quotes, picks one to award.
Menu / route	RFQ detail · Quotes tab · 'Shortlist' / 'Award' buttons
Available to roles	buyer
Backend API	PUT /api/rfqs/:id (status=SHORTLISTED) · POST /api/rfqs/:id/award (quoteId)

CLICK-BY-CLICK

- CLICK** Click 'Shortlist' on the quotes you want to consider
→ Status=SHORTLISTED
- CLICK** Click 'Award' on the winning quote
→ Status=AWARDED · awardedQuoteId set
- WAIT** Auto-conversion
→ audit-requests-master row created with rfqId link · RFQ status=CONVERTED

5. Full lifecycle walkthrough — one transaction, end-to-end

One RFQ from draft to award + auto-convert.

#	Persona	State transition	Action	API	Observed	Result
1	Priya Buyer	— → DRAFT	+ New RFQ · supplier · site · products · closingAt=+30d	POST /api/rfqs	rfqNumber=RFQ-XXXXXX · status=DRAFT	PASS

Expected DB state after this step:

```
audit-rfqs { _id, rfqNumber, supplierOrgId, siteId, productIds, closingAt, status: 'DRAFT', auditTrail: [] }
```

#	Persona	State transition	Action	API	Observed	Result										
 Audits New Audit CAPAs Documents Deviations Complaints Changes Training Equipment Risk Reviews Buyer RFQs Audit RFQs Create, invite auditors, compare quotes, and award audits. New RFQ Draft Published Closing Soon Awarded <table><tr><th>RFQ</th><th>Title</th><th>Status</th><th>Closing</th><th>Action</th></tr><tr><td colspan="5">No RFQs found for this filter.</td></tr></table> state-screens/rfq-list.png							RFQ	Title	Status	Closing	Action	No RFQs found for this filter.				
RFQ	Title	Status	Closing	Action												
No RFQs found for this filter.																
2	Priya Buyer	DRAFT → PUBLISHED	POST /publish	POST /api/rfqs/:id/publish	Status flips	PASS										
3	Priya Buyer	PUBLISHED → PUBLISHED (with invites)	POST /invite · auditorOrgIds[]	POST /api/rfqs/:id/invite	invitedAuditors[] populated	PASS										
4	Maria (auditor org) auditor	– → (quote) SUBMITTED	POST /quotes · lineItems · totals · schedule	POST /api/rfqs/:id/quotes	Quote row with status=SUBMITTED	PASS										
Expected DB state after this step:																
audit-rfq-quotes { _id, rfqId, auditorOrgId, lineItems, totals, proposedSchedule, status: 'SUBMITTED' }																
5	(system) —	PUBLISHED → QUOTES_RECEIVED	Auto-fired on first quote submission	—	RFQ status auto- flips to QUOTES_RECEIVED	PASS										
6	Priya Buyer	QUOTES_RECEIVED → AWARDED	POST /award · quoteId=winner	POST /api/rfqs/:id/award	Status=AWARDED · awardedQuoteId set	PASS										
Expected DB state after this step:																
audit-rfqs { status: 'AWARDED', awardedQuoteId: maria-quote._id }																
7	(system) —	AWARDED → CONVERTED	Auto-create audit- requests-master row with rfqlId link	(auto)	audit-requests- master row created · RFQ status=CONVERTED	PASS										

state-screens/rfq-list.png

#	Persona	State transition	Action	API	Observed	Result
Expected DB state after this step:						
audit-requests-master { rfqId, supplierId, scope, phaseState: { currentPhase: 'INITIATED' } }						

6. AI assistance map

Each AI assist is a separate endpoint. Outputs flow through the grounded-generation runtime (citation gate, confidence floor, schema validation, accept/reject audit trail).

AI agent	Attached to state(s)	Endpoint	Where in UI	What it returns	Provider
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7. Regulator-trace matrix

For each lifecycle state, the specific clauses + evidence Hawkeye captures so an inspector can audit the record. Maps to FDA / ISO / ICH / EU GMP source clauses.

State	Cited clauses	Evidence captured	Backing records
PUBLISHED	–	title + scope + invitedAuditors[]	audit-rfqs
AWARDED	–	awardedQuoteld + buyer signature on award	audit-rfqs + audit-rfq-quotes
CONVERTED	ISO 19011 §5.3	rfqId link from new audit-request to source RFQ	audit-requests-master + audit-rfqs

8. Test results matrix

Suite	Scope	Result	Evidence
novex-walkthrough.spec.ts	/rfqs + /request-audit UI render	PASS	09-priya.png
(deferred) eqms-lifecycle.spec.ts · rfq	Full RFQ lifecycle requires buyer + auditor org orchestration	SKIP	Not yet automated

9. Known gaps + roadmap

- Full RFQ E2E lifecycle test — buyer + auditor orchestration spec. MEDIUM
- Auto-EXPIRE on closingAt — Cron flips status to EXPIRED when closing date passes without award. MEDIUM
- AI quote-comparison helper — Side-by-side quote comparison with line-item delta + recommendation. LOW